



Hope Foundation's
International Institute of Information Technology (I²IT), Pune
DEPARTMENT OF COMPUTER ENGINEERINGS
Academic Year 2026-27 Semester I

THEORY QUESTION BANK

Class TE (2024) - C

Course Name: Computer Networks

Que No.	Question	BTL	Marks
Unit 1 (CO302.1)			
1	Describe PAN, LAN, MAN, and WAN with neat diagram.	BTL-2	5
2	Describe the protocol hierarchy and explain the importance of layered architecture.	BTL-2	5
3	Explain the design issues for layers.	BTL-2	5
4	Differentiate between connection-oriented and connection-less services.	BTL-2	5
5	Explain service primitives with a suitable example.	BTL-2	5
6	Compare the OSI Reference Model and TCP/IP Reference Model.	BTL-3	5
7	Draw and explain the OSI Reference Model. State the functions of each layer.	BTL-3	5
8	Draw and explain the TCP/IP Reference Model. State the functions of each layer.	BTL-3	5
9	Write short note on guided and unguided transmission media.	BTL-3	5
10	Explain Ring, Star, Bus and Mesh topologies with neat diagram and list advantages and disadvantages.	BTL-2	5

Unit 2 (CO302.2)			
1	Explain Framing, Error control and Flow control	BTL-2	5
2	Justify answer using CRC for divisor: 1101 dividend: 100100	BTL-4	5
3	Sender Sends code 1001101 justify receiver using Hamming code.	BTL-4	5
4	Illustrate the Stop-and-Wait Protocol using a neat timing diagram.	BTL-3	5
5	Describes design issues in DLL	BTL-2	5
6	Explain types of variable frame.	BTL-2	5
7	Demonstrate the working of the Go-Back-N protocol with an example.	BTL-3	5
8	Demonstrate the working of the Selective Repeat protocol with an example.	BTL-3	5
9	Explain Packet over SONET (PoS) technology and its applications in communication networks.	BTL-2	5
10	Explain parity and checksum techniques for error detection with suitable example.	BTL-2	5